

Concept Question 3.1 Stellar dynamics

Consider three stars in the Milky Way's disk, at distances of 20 000, 25 000 and 30 000 light-years from the Galactic center, and suppose that right now all three are lined up in such a way that it is possible to draw a straight line through them to the center of the Galaxy. How will the relative positions of these three stars change over time? Suppose that their orbits are all circular and in the plane of the disc.

Problem 3.2 Galactic rotation

In this exercise you will explore radial velocity mapping under different rotation models.

- a) Calculate and draw the $v_r(\ell)$ map of the galacto-centric gas rings at $R = 3, 5, 8, 12$, and 16 kpc.
Assume: $R_0 = 8$ kpc, $\Theta_0 = 200$ km/s and a flat rotation curve $\Theta(R) = \Theta_0$.
- b) Determine the curve for maximum positive radial velocity as a function of ℓ for $0^\circ < \ell < 90^\circ$ and for the same parameters as above: $R_0 = 8$ kpc, $\Theta_0 = 200$ km/s and $\Theta(R) = \Theta_0$.
- c) Assuming a Keplerian disk, calculate and draw the $v_r(\ell)$ map of the galacto-centric gas rings at $R = 3, 5, 8, 12$, and 16 kpc. Determine again the curve for the maximum positive radial velocity for $0^\circ < \ell < 90^\circ$ for the solution of a Keplerian disk and compare this result with the result from sub-question b). What is the expected velocity difference for the two maximum radial velocities as function of ℓ ?